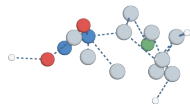


a Composition to 3D coordinates

Input: atomic types, charge and spin
Harmonic source



$$X_0 \sim q_0(\cdot | c)$$

X, t, c

Frozen parent prediction

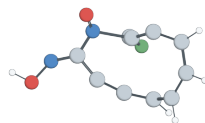
H

Learned correction

$$v^{\text{corr}} = v + \eta A_\phi$$

Integrate

+



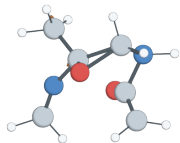
$$X_{\text{gen}} \in \mathbb{R}^{20 \times 3}$$



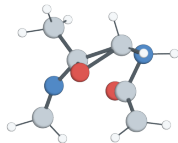
Charge 0; spin multiplicity 1

Generated structure

b Physical supervision at predicted endpoints



Force
shift



Supervise

Symmetric pair features
Bounded scalars
Opposite pairwise vectors

$$A_\phi \approx \delta(H)/(1 - p)$$

Provisional endpoint H

Target $H + \delta(H)$

Forces are used during training; sampling uses neural networks.